

ESE 2180 Project One: Solving Linear Circuits

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1 Project Information

This project will provide familiarity with techniques for solving matrix equations by exploring a case study in linear circuits. The project is due on **September 29**. You may work in groups of up to **two**. You will turn in a lab report and MATLAB source code as described in Section 8.

2 Learning Outcomes

The learning outcomes of this project are:

- Formulate a real-world physical problem as a system of linear equations
- Write code to solve a linear system using Gaussian elimination
- Prepare a report and well-commented code

3 Resistor Networks

The electric voltage at a particular point refers to the electrical potential energy at that point, typically relative to a ground state that is assumed to have zero voltage. Ohm's Law gives a linear relationship between the voltage at two points A and B , denoted V_A and V_B respectively, and the current I_{AB} from A to B . That relationship is given by $V_A - V_B = I_{AB}R_{AB}$, where R_{AB} is the electrical resistance of the connector between A and B (Fig. 1). An equivalent relationship is given by $I_{AB} = C_{AB}(V_A - V_B)$, where $C_{AB} = 1/R_{AB}$ is the conductance between A and B .

A resistor network is a collection of points (or nodes) $n_1, \dots, n_M$ that are connected by resistors (Fig. 2), where M is the number of nodes. Some of these nodes may be voltage sources, i.e., they are fixed to a particular voltage. By convention, we choose one node as a *ground* that has zero voltage. The voltages of the remaining nodes are determined by the Ohm's Law and the Kirchhoff Voltage and Current Laws [1] (KVL and KCL), which are defined by

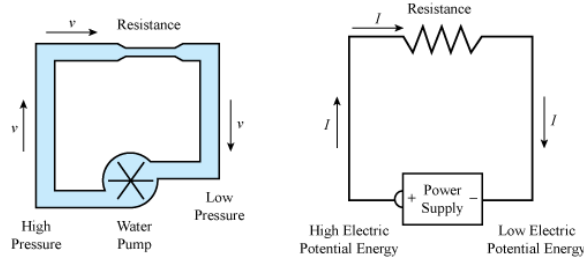


Figure 1: Ohm's Law illustration

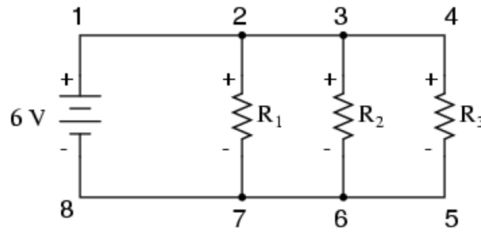


Figure 2: Resistor Networks

- Kirchhoff Voltage Law (KVL) – The sum of the changes in voltage around any loop in the network is equal to zero. In other words, if $(A_1, A_2), (A_2, A_3), \dots, (A_r, A_1)$ is a sequence of links in the network, then

$$\sum_{i=1}^{r-1} (V_{i+1} - V_i) + (V_1 - V_r) = 0. \quad (1)$$

- Kirchhoff Current Law (KCL) – The sum of the net currents entering any junction is equal to zero. In other words, if $A_1, \dots, A_r$ is the set of nodes that are connected to node B , then

$$\sum_{j=1}^r I_{A_j B} = 0 \quad (2)$$

4 Solving Circuits Using Node Voltages

The main idea of node voltages is to determine the voltages at each node in the graph using KCL (Eq. (2)). Each node that is not a voltage source results in one linear equation. The following procedure is used to solve linear circuits using node voltages:

1. Identify the nodes in the graph. For each node i that acts as a voltage source, add an equation $V_i = u_i$, where u_i is the voltage assigned to i .

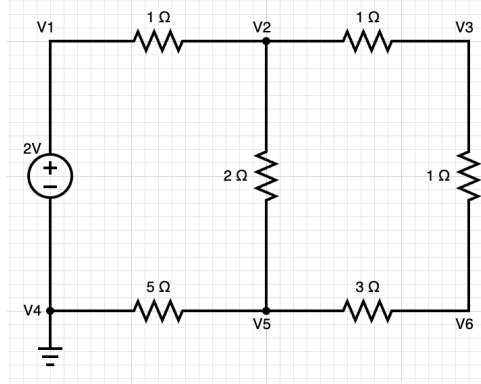


Figure 3: Example of node voltage analysis.

2. For each remaining node, write the KCL equation. For a node B, this will be of the form (2).
3. Convert the currents in the KCL equation to their corresponding voltages using Ohm's Law, so that (2) becomes

$$\sum_{j=1}^r \frac{1}{R_{A_j B}} (V_{A_j} - V_B) = 0 \quad (3)$$

Consider the example shown in Fig. 3. There are six nodes. Solving for the node voltages, we obtain the following system of equations:

For **Node** V_2 , we apply KCL:

$$\frac{V_2 - 2}{1} + \frac{V_2 - V_3}{1} + \frac{V_2 - V_5}{2} = 0$$

Multiplying through by 2 to clear the fraction and simplifying, we get:

$$5V_2 - 2V_3 - V_5 = 4$$

For **Node** V_3 , the KCL equation is:

$$\frac{V_3 - V_2}{1} + \frac{V_3 - V_6}{1} = 0$$

Simplifying, this gives:

$$-V_2 + 2V_3 - V_6 = 0$$

For **Node** V_5 , we have:

$$\frac{V_5 - V_2}{2} + \frac{V_5}{5} + \frac{V_5 - V_6}{3} = 0$$

Multiplying through by 30 and simplifying:

$$31V_5 - 15V_2 - 10V_6 = 0$$

Finally, for **Node** V_6 , the KCL equation is:

$$\frac{V_6 - V_3}{1} + \frac{V_6 - V_5}{3} = 0$$

Simplifying, we get:

$$4V_6 - 3V_3 - V_5 = 0$$

Next, we write these equations in matrix form:

$$\begin{pmatrix} 5 & -2 & -1 & 0 \\ -1 & 2 & 0 & -1 \\ -15 & 0 & 31 & -10 \\ 0 & -3 & -1 & 4 \end{pmatrix} \begin{pmatrix} V_2 \\ V_3 \\ V_5 \\ V_6 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Solving this system, the node voltages are found to be:

$$V_2 = 1.73 \text{ V}, \quad V_3 = 1.65 \text{ V}, \quad V_5 = 1.35 \text{ V}, \quad \text{and} \quad V_6 = 1.58 \text{ V}$$

5 Effective Resistance

The *effective resistance* between two nodes i and j is defined as follows. Suppose that 1 A of current is injected into node i and 1 A of current is extracted from node j , with no external current injected into any other node. We then solve the circuit for the node voltages v and define

$$R_{\text{eff}}(i, j) = v(i) - v(j).$$

6 Your Assignment

For the following assignments, you will work with the 5×5 resistor network shown in Fig. 4. The nodes are numbered row by row from 1 to 25. The file `node_resistances.txt` specifies the resistance of each link, and the file `node_voltages.txt` specifies the fixed node voltages. In the provided network, Node 1 is fixed at 10 V and Node 25 is fixed at 0 V. Please complete the following steps.

1. Let $\mathcal{N}(i)$ denote the set of nodes connected to node i , and let R_{ij} denote the resistance between connected nodes i and j . For the 5×5 resistor network, derive the system of equations

$$Ax = b$$

for the node voltages. Specify A_{ij} and b_i for $i, j = 1, \dots, 25$.

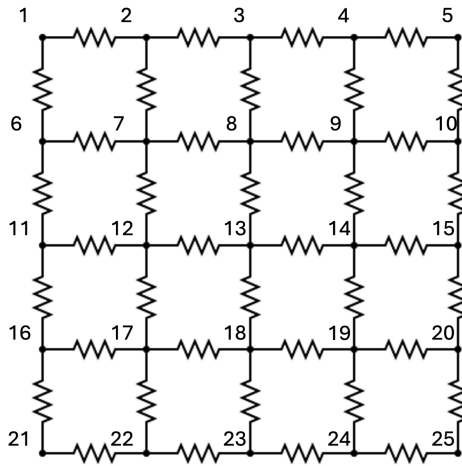


Figure 4: The 5×5 resistor network used in this project. Nodes are numbered row by row from 1 to 25.

2. Write a function that reads the resistance data from `node_resistances.txt`. Each line of the file has the form

```
NODE1    NODE2    RESISTANCE
```

where the first two entries identify the two connected nodes and the third entry gives the resistance of the link. For example, if there is a resistance of 5 between nodes 2 and 3, then the corresponding line would be

```
2 3 5
```

3. Write a function that reads the prescribed node voltages from `node_voltages.txt`. Each line has the form

```
NODE      VOLTAGE
```

For example, if the voltage at node 3 is equal to 6.2, the corresponding line would be

```
3 6.2
```

4. Using the resistance and voltage data read in Parts 2 and 3, construct the matrix A and vector b for the 5×5 network.
5. Write functions that compute the LU factorization of a given matrix and solve lower- and upper-triangular systems.

6. Use the functions that you wrote to solve for the node voltages and compute the current through each link.
7. Write the computed node voltages and currents through each link to an output file.
8. Analyze the behavior of grid resistor networks as follows:
 - (i) Consider the 5×5 grid with the top-left node fixed at 10 V and the bottom-right node fixed at 0 V. Suppose that all links have the same resistance R . The total power dissipated in the network is

$$P = \sum_{(i,j) \in E} I_{ij}(V_i - V_j).$$

Plot the total power dissipated as a function of R for $R = 1, 2, \dots, 20 \, \Omega$.

- (ii) Suppose that every link has resistance $1 \, \Omega$, the top-left node is fixed at 10 V, and the bottom-right node is fixed at 0 V. Plot the current entering the bottom-right node for $k \times k$ grid networks using the same grid topology and row-by-row node numbering convention with

$$k = 5, 10, 15, 20, 25.$$

9. Write a function that computes the effective resistance $R_{\text{eff}}(i, j)$ between any specified pair of nodes i and j in a resistor network.
10. Consider a 5×5 grid in which every link has the same resistance R . Plot the effective resistance between the top-left and bottom-right nodes as a function of R for $R = 1, 2, \dots, 20 \, \Omega$.

7 Writing a Lab Report

A lab report is a written document describing the steps you took to complete the project and your results. The report should include the following sections:

- Introduction – Briefly describe the motivation and goals of the project.
- Background – Describe needed background, for example, regarding Kirchhoff's laws.
- Methods – Describe the methods you took to solve the problem. For example, how you derived the system of linear equations for solving the circuit, how you implemented the LU factorization in your code, and how you computed the effective resistance.
- Results – Should contain the plots as described above.
- Conclusions – Conclusions of your study and any steps for future work.

8 Deliverables

Please submit the following via Canvas:

- A project report (one per group) as described in Section 7.
- Your MATLAB source code, including all functions needed to reproduce the results in the report.

References

- [1] C. K. Alexander, M. N. Sadiku, and M. Sadiku, *Fundamentals of electric circuits*. McGraw-Hill Higher Education Boston, MA, USA, 2007.